

Preferences Become Measurable at Supervised Fine-Tuning

Phase 3 Results: Staged Attribution of Risk and Framing Behavior Across Post-Training

J. Blakeslee

RAND School of Public Policy

AI Sprint Practicum (Course 1012)

September 2026

1 Introduction

This project treats large language models as economic agents and recovers their preference primitives from their choices. Phase 1 fit a structural discrete-choice model to 1,200 controlled lottery choices from `claude-haiku-4-5` and found that the model prices the probability of winning and nearly ignores the size of the prize, with an extreme S-shaped Prelec weighting function ($\gamma = 2.69$). Phase 2 held terminal wealth fixed and varied only the wording, and found a violation of frame invariance: 40% versus 10% gamble-taking on arithmetically identical lotteries, with a stated-reference loss-aversion estimate of $\lambda = 3.234$ under $\phi = 1$. Both phases measured a single closed frontier model, and both ended on the same open question. Where do these properties come from?

Phase 3 takes it up by moving to open-weight models whose training stages are public. The identical Phase 1 and Phase 2 instruments are carried to matched base and post-trained checkpoints from three families, and choice probabilities are read directly from next-token logits rather than sampled. The decisive asset is `OLMo-2-7B`, which ships four public checkpoints along one training path: base, supervised fine-tuned (SFT), preference-optimized (DPO), and final Instruct. Running the same instruments across that staircase, in a single elicitation format, turns a coarse base-versus-instruct contrast into a stage-level attribution.

The headline is model-free. Supervised fine-tuning is the largest single step along the `OLMo-2` training path in both quantities that need no preference model. From base to final Instruct, dominance compliance rises from 0.555 to 0.896 and the frame gap from +0.099 to +0.349. SFT accounts for about three fifths of each change (dominance +0.200, 95% CI [+0.177, +0.226]; frame gap +0.144, CI [+0.133, +0.155]). The later stages are still active. DPO adds +0.126 to dominance compliance and +0.075 to the frame gap, both clearly different from zero, and the final polish adds a small further amount.

The structural reading of the same staircase is less settled than the model-free one, and we report why. The Phase 2 specification has no constant term. Adding one, a fixed lean toward or away from the gamble regardless of payoffs, improves the fit at every `OLMo` stage, and it changes what the parameters say. Without the constant, SFT moves loss aversion from zero to $\lambda = 1.14$ and the parameters then hold nearly still. With it, the post-trained models put no weight on the loss side of a mixed gamble (λ at zero at every stage), and what moves at SFT is the lean itself, from toward the gamble (+0.77) to toward the safe option (−0.80). The lean keeps deepening through

DPO (-1.02) and the final stage (-1.30), which accounts for part of what the later stages change. A simulation shows the design separates the two specifications, so the data favor the second. The paired comparisons say the same without a model. The models gamble more, not less, when a lottery is described as a possible loss, and that extra gambling grows with the size of the stated gain and barely responds to the size of the stated loss. The framing response is therefore not loss aversion, whatever else it is. One structural result survives both specifications: after SFT the preference parameters change by less than a tenth of the SFT step.

Where the profile comes from is a separate question, and the staircase answers it only in part. The OLMo base model places 0.555 of its probability on the larger of two sure amounts, close to a coin flip, so SFT is the first stage at which the profile can be measured at all. These data cannot distinguish a profile that SFT installs from one planted in pretraining that SFT makes visible, which is the account Itzhak et al. (2025) give for cognitive biases generally. What the staircase does establish is order. The measurable economic profile is in place before preference optimization begins, and the structural parameters that summarize it, under either specification, change little afterward. Two further results hold across families. Base models barely track value at all, consistent with the poor utility fits Wang et al. (2025) report for the same Llama and Qwen pairs, and the dominance control shows why those fits fail. And the framing response is family-specific: Haiku suppresses gambling under loss wording, Qwen amplifies it sharply, OLMo amplifies it moderately, and Llama barely reacts.

Everything above comes from an immediate one-letter answer, and a reason-then-answer arm, run after the reviewer’s induced-valuation result, shows how much that restriction matters. Allowed to write out their reasoning before answering, the models follow an instructed expected-value rule on 0.965 of clear-cut cells (Qwen Instruct) and 0.803 (OLMo Instruct), against roughly 0.55 when answering immediately, and uninstructed they compute expected value on their own in most traces. The framing response changes with the regime. Qwen’s strong amplification under loss wording ($+0.626$) reverses to a small suppression (-0.135 , 95% CI $[-0.221, -0.053]$), produced by one recurring accounting error: the model compares the gamble’s expected change in wealth with the sure option’s total. The staircase changes too. Under reasoning, OLMo SFT keeps a large frame gap ($+0.375$) that is gone by the DPO checkpoint ($+0.015$; difference -0.360 , 95% CI $[-0.429, -0.287]$), and the final stage adds nothing detectable. Preference optimization, invisible in the immediate-answer parameters, sets how the model behaves when it deliberates, and the attribution to supervised fine-tuning is a result about the immediate-answer regime. The traces also carry a caution for anyone who reads reasoning as a record of the computation: most of Qwen’s remaining misses outside the loss frame compute the expected value correctly and then misstate the comparison in favor of the safe option, and a prefill test traces the misstatements to one stock phrase. Two follow-ups sharpen the picture. One added sentence, telling the model to compare final dollar amounts, removes Qwen’s reasoning-regime framing error entirely (mixed-frame accuracy 0.869 to 1.000), so that error is a bookkeeping habit, not a preference; OLMo Instruct makes the same slip and ignores the sentence (0.588 to 0.581). And paying the model through training, the closest available analogue to Smith’s payment, does not teach the computation in the immediate-answer regime: across six configurations the paid model settled on a fixed answer and never learned to tell better gambles from worse ones.

2 The Result in Plain Language

An AI model is built in stages. First it is trained on a very large amount of text, producing a base model: a system that continues text plausibly and has never been taught to answer questions. Then it is fine-tuned, in several distinct steps, to behave like a helpful assistant. Most published work on AI decision-making tests only the finished assistant, which cannot tell you whether a quirk came from reading the internet or from the training that came after. One model family, OLMo-2, publishes a snapshot after each training step. So we asked the money questions from our earlier phases to the same model at four points in its life, in identical wording each time, and watched what changed.

The biggest change came at the first step. The base model was close to incoherent as an economic decision-maker. Offered a sure \$50 against a sure \$70, no gambling involved, it picked the larger amount barely more often than a coin flip. The base model does not have odd preferences; it has no preferences in any usable sense, and is producing a plausible-looking letter rather than comparing two amounts of money. After supervised fine-tuning, the first and simplest of the assistant-training steps, it picked the larger amount three quarters of the time and began reacting to how the choices were worded. The two later steps, the ones built on human preference comparisons, kept pushing in the same direction: the model became more reliable on the easy questions, somewhat more sensitive to wording, and steadily more inclined toward the safe option across the board. The wording effect is not the human one, though. People avoid a gamble when its downside is called a loss; these models take it more often, and they respond to how big the stated gain is, not to how big the stated loss is. The later stages are the ones people usually blame, since human feedback enters there. Here the measurable preferences are already in place before those stages begin. Whether supervised fine-tuning created them, or only made visible something the model absorbed from its reading, is a question the base model’s near-random answers leave open.

Three other things came out of the same runs. The wording effect from Phase 2 has no single direction across systems. Claude Haiku became much more cautious when the downside was called a loss. Qwen became dramatically more willing to gamble under the same wording change. OLMo also became more willing, less dramatically. Llama hardly noticed. Four systems, four responses to one manipulation. The effect is a property of how a particular model was trained rather than of AI systems in general, so anyone reporting a framing bias from one model and calling it a fact about language models is reporting a fact about that model.

The answer also depends on how you ask, in a way that is model-specific and easy to miss. We put the same questions to each finished assistant in two standard formats. Qwen behaved sensibly in the chat format and collapsed into near-total refusal to gamble in the other. OLMo did the reverse. Llama was roughly stable in both. A study comparing two models in one fixed format may be comparing an engaged model against one that has checked out. The check is cheap: before analyzing anything, confirm the model reliably picks \$70 over \$50. If it does not, the rest of its answers are not preferences.

The third came from our reviewer, and it is the most uncomfortable. Experimental economists have a standard move: rather than guess what a subject wants, tell the subject what to want and pay them for it. We ran the instruction-only version, telling the models in plain language to be expected-value maximizers, to multiply payoff by probability and take the larger number. They could not do it. Compliance with the stated rule never rose above roughly 55% at any training stage, close to a coin flip, while the same models followed the part of the instruction requiring no arithmetic almost perfectly, reaching 100% on the easy questions. Asked for a harder rule involving a square

root, one model’s compliance fell to 12% with a legible error pattern: it compared the transformed payoffs and dropped the probability entirely. So we let them work the problem out in writing before answering. Qwen then followed the rule 96% of the time and OLMo 80%. The obstacle was the one-letter format. With room to reason, the models’ reaction to loss wording changed as well, and Qwen’s reversed. Working through the arithmetic, it made the same bookkeeping mistake again and again, counting the gamble as a change in its money and the sure option as a total, and that mistake always favors the safe choice. Adding one sentence, “compare the final dollar amounts,” made the mistake disappear. We also tried paying the model for good choices by training it on the lottery payoffs. When it had to answer at once, payment did not teach it the arithmetic; it only learned to give the same answer every time.

3 Experimental Design

3.1 Instruments, grid, and subjects

Phase 3 reuses the Phase 1 and Phase 2 instruments without modification. The exported grid (`data/phase3_grid.csv`) holds 1,680 design cells: 400 Phase-1 risk cells (40 procedurally generated contamination-safe gambles $\times$ 5 paraphrase templates $\times$ 2 counterbalanced orders), 1,200 Phase-2 framing cells (the same 40 gambles $\times$ 3 frames $\times$ 5 templates $\times$ 2 orders), and 80 dominant controls pitting a sure amount against a strictly larger sure amount. Every model sees byte-identical prompts, so nothing in the cross-model comparison is a difference in the instrument.

Three families, seventeen runs, 28,560 elicited choice probabilities. Qwen2.5-7B (base, Instruct under both formats); Llama-3.1-8B (base, Instruct under both formats), run once Hugging Face access to the gated repository was approved; and OLMo-2-1124-7B in four stages (base, SFT, DPO, Instruct) plus Instruct under the chat format. Six further induced runs cover the OLMo staircase under risk-neutral induction and Qwen Instruct chat under both induced rules. Six reason-then-answer runs (Section 3.4) cover Qwen Instruct chat and OLMo SFT and Instruct few-shot, each uninduced and induced.

3.2 Elicitation by deterministic logits

Base models do not follow instructions, so the Phase 1–2 forced-tool-choice harness cannot run on them. Phase 3 reads the choice probability directly. Each prompt is wrapped in a $k = 4$ few-shot completion (exemplar answers balanced two A and two B, so no letter prior is induced) or passed through the model’s own chat template, and the full next-token distribution is read at the answer position:

$$P(\text{gamble}) = \frac{\sum_{t \in \mathcal{T}_g} P(t)}{\sum_{t \in \mathcal{T}_A} P(t) + \sum_{t \in \mathcal{T}_B} P(t)},$$

where $\mathcal{T}_A$ and $\mathcal{T}_B$ are the candidate answer tokens resolved per tokenizer, with and without a leading space. The unnormalized A/B mass $\sum_{\mathcal{T}_A} P + \sum_{\mathcal{T}_B} P$ is logged per cell as a compliance diagnostic and cells below 0.20 are excluded. One forward pass per cell replaces roughly 25 sampled repetitions, because the probability is the quantity the logit likelihood wants. Runs use 4-bit quantization on a free Colab T4.

This also settles a Phase 1–2 nuisance. Temperature 1.0 sampling was treated there as a choice-error process, an assumption the earlier phases could not test. Phase 3 involves no sampling anywhere:

the logits are deterministic given the prompt, and all randomness in the design is the randomization the experimenter controls, namely templates, orders, and the payoff grid.

A base and instruct pair differ in weights and in natural elicitation format at the same time, so a parameter gap could be an artifact of format. Every instruct model is therefore run under both formats, its chat template and the identical raw few-shot completion the base model receives. This control was mandatory by design, and it earned its place (Section 5.4).

3.3 The induced-valuation arm

Following the reviewer’s first point (Section 4.3), a second arm asserts a utility function instead of eliciting one. The same 1,680 cells are re-run with an instruction prepended: either risk-neutral expected-value maximization, or expected utility with $u(x) = \sqrt{x}$. Compliance with the induced optimum is scored by `analysis/score_induced.py` as the probability mass placed on the induced-optimal option, reported on cells where the induced rule clearly separates the two options ($|\ln(\text{EU}_g/\text{EU}_s)| \geq 0.05$) and separately on the dominant controls, where both rules agree and no arithmetic is required. This is the instruction-only version of Smith’s (1976) induced-value procedure, without the incentive that makes the original work.

3.4 The reason-then-answer arm

A failure of the induced arm could mean the models cannot compute in an immediate-answer format or that they will not adopt the stated rule. A reason-then-answer arm separates the two. The model generates reasoning greedily, with no sampling and at most 320 tokens, until it writes its own **Answer:** line. The reasoning is kept, **Answer:** is appended, and the A/B distribution is read at that position exactly as in the main arm, so readout and scoring are unchanged. In the chat format the prompt asks the model to think the choice through and end with its answer. In the few-shot format each of the four exemplars gains a one-line reasoning step that states the certain amount or the probability times the payoff for each option, without stating a decision rule. Those exemplars demonstrate the expected-value arithmetic, so uninduced few-shot reasoning is partly primed toward it, and the cleanest comparisons are within the arm.

Reasoning costs roughly ten times as much compute as an immediate answer, so the arm covers 20 of the 40 gambles per instrument, with every frame, template and order of each (840 cells, fixed seed). All comparisons with the immediate-answer runs below are on these matched cells. Subjects are each model in the format where it passes the dominance control: Qwen Instruct in chat, OLMo SFT and OLMo Instruct in few-shot, each uninduced and under the risk-neutral instruction, and OLMo DPO in few-shot to locate the reasoning-regime change between training stages, and Llama Instruct in chat as a third family.

4 Econometric Model and Estimation

4.1 Fractional maximum likelihood

The observed data are probabilities, one per design cell, rather than Bernoulli draws. The Phase 1 and Phase 2 likelihoods carry over unchanged if the Bernoulli log-likelihood is replaced by the binomial deviance with the observed probability as the target, the standard fractional-response

objective (Papke and Wooldridge, 1996):

$$D(\theta) = \sum_c \left[y_c \ln \frac{y_c}{\hat{p}_c(\theta)} + (1 - y_c) \ln \frac{1 - y_c}{1 - \hat{p}_c(\theta)} \right],$$

where y_c is the elicited probability of the gamble in cell c and $\hat{p}_c(\theta)$ is the structural choice probability. The structural model is the Phase 2 specification: reference-dependent value $v(x) = (x - \rho)^\alpha$ for $x \geq \rho$ and $-\lambda(\rho - x)^\alpha$ below, with $\rho = \phi \cdot \text{anchor}$, Prelec weighting $w(p) = \exp(-(-\ln p)^\gamma)$, a Fechner noise term μ , and the position nuisance parameter δ . The Phase 1 risk cells are the anchor-zero rows of the same likelihood, so one estimator covers both instruments. Fitting is by multi-start Nelder-Mead under the Phase 2 stated-reference restriction $\phi = 1$. Parameter intervals are profile-likelihood intervals from a refit with wide bounds ($\lambda \in [10^{-4}, 8, 103]$, $\gamma \in [0.018, 54.6]$, $\alpha \in (0, 1)$): each of λ , γ and α is fixed on a grid, the remaining parameters are re-optimized, and the interval is read where the scaled profile crosses the χ^2_1 95% value. Because the objective is a quasi-likelihood and the ten presentations of a gamble (five templates, two orders) are not independent, the profile is scaled by a sandwich variance clustered on the 80 gambles. Where a parameter sits at the edge of the wide box the sandwich is invalid, and the scaling falls back to Pearson dispersion, which ignores the clustering and gives intervals that are probably too narrow. A contrast between two subjects combines their two profiles exactly, since they share no parameters. An earlier version of this phase used a bootstrap over design cells; it failed at the old parameter bounds, where one interval overflowed to -3.8×10^{48} .

Two checks on these intervals and on the specification are reported in Section 5.4. A wild cluster bootstrap around each fitted model (`analysis/bootstrap_refit.py`; Rademacher weights on each gamble’s block of residuals, 400 replicates, the same weights for every subject so that stage contrasts are paired) gives every subject, boundary or not, an interval calibrated the same way (Cameron et al., 2008). At a boundary estimate the bootstrap is itself inconsistent (Andrews, 2000), so those intervals are read as a check on the dispersion fallback rather than as exact coverage. And a second specification adds a constant c to the choice index, with template and gamble random intercepts on top of it (`analysis/template_effects.py`); random-effect variances are tested against zero as boundary hypotheses (Self and Liang, 1987).

Validation gate: the estimator recovers the generating parameters near-exactly on the simulated probability agent (`data/phase3_dryrun.csv`), so the degenerate fits reported below are properties of the models’ choice patterns rather than estimator failures.

4.2 Model-free quantities and held-out validation

Two quantities need no structural model and are reported as primary throughout. **Dominance compliance** is the mean probability placed on the strictly larger sure amount across the 80 dominant controls, a per-model, per-format engagement check: a subject that fails it is not expressing preferences. **The frame gap** is mean $P(\text{gamble})$ in the mixed frame minus the gain frame over terminal-wealth-identical lotteries, the Phase 2 headline in probability units. Structural parameters are reported alongside, under both specifications, for the reasons given in Section 5.4 and Section 8.

Phase 1 closed on an honest weakness: the structural model essentially tied a probability-only logit on in-sample fit and lost on BIC. `analysis/holdout_validation.py` settles it out of sample. The 40 gamble identifiers are split 30/10, each candidate is fit on the training gambles, and mean per-cell held-out deviance is evaluated on cells belonging to gambles the fit never saw, over 10 random splits. Candidates are the structural model, a two-parameter logit in the win probability alone,

and the training-mean constant.

4.3 Reviewer feedback and design response

Philip Armour reviewed the Phase 1–2 design memo and returned four points in August 2026. Each is answered by an element of the design above, and the correspondence is worth stating explicitly because three of the four changed what this phase reports.

1. **Induced valuation.** His central point invoked the induced-value tradition: if a stated utility function cannot be induced in the subject, elicited “natural” preferences rest on weaker ground, and if it can, compliance calibrates how far to trust the elicitation. This became the arm in Section 3.4, and it lands harder here than in Phase 1, because the OLMo staircase can ask where in post-training a model becomes the kind of subject induced-value theory presumes. His harder variant, denominating payoffs in a token or computation budget so the stakes are real, is the fall workplan.
2. **The λ - ϕ collinearity.** He pointed to Rees-Jones and Wang (2022) on testing reference points, a minimally parametric approach to the exact degeneracy Phase 2 hit. It is the reading gate for the Phase 2 revision.
3. **Temperature and what is being randomized.** He asked whether the model’s outputs are deterministic given the same prompt. They are, and the Phase 3 readout uses precisely that: no sampling anywhere, with the design randomization (templates, orders, payoff grid) supplying the distribution the bootstrap resamples. His further ask, sampling decision-maker characteristics from CPS or ACS distributions, changes the estimand from the deployed assistant to a simulated population, so it belongs as an additional arm rather than a replacement.
4. **Out-of-sample prediction.** An emphatic yes, and pure analysis on data already in hand. Implemented as Section 4.2; it answers the Phase 1 near-tie (Section 5.10).

The project sits inside the structural behavioral economics tradition DellaVigna (2018) surveys: estimate the parameters of a behavioral model rather than demonstrate a bias, so magnitudes are comparable across designs and the identifying assumptions are stated. That is what makes a stage-level attribution meaningful rather than decorative.

5 Results

5.1 Data inventory and engagement

All eleven uninduced runs completed with a 0% exclusion rate and high A/B token mass (medians 0.908 to 1.000), so no model is failing to treat the prompt as an A/B question. The induced runs excluded at most 1.8% of cells. Table 1 reports the inventory alongside the two model-free quantities, with the Phase 2 Haiku result as the reference row.

5.2 Base models barely track value

All three base models emit answer tokens with high confidence and then fail the easiest question in the instrument. Dominance compliance is 0.555 for OLMo, 0.623 for Llama, and 0.639 for Qwen,

Table 1: Data inventory and model-free quantities. Each run scores the same 1,680-cell grid. A/B mass is the median unnormalized probability on the answer tokens. Dominance is mean probability on the strictly larger sure amount over 80 controls. The frame gap is mixed minus gain uptake on terminal-wealth-identical lotteries.

Subject	Format	A/B mass	Dominance	Gain	Mixed	Frame gap
Qwen2.5-7B base	fewshot	0.960	0.639	0.161	0.365	+0.203
Qwen2.5-7B Instruct	fewshot	0.995	0.680	0.000	0.001	+0.001
Qwen2.5-7B Instruct	chat	1.000	0.927	0.267	0.893	+0.626
OLMo-2-7B base	fewshot	0.908	0.555	0.499	0.598	+0.099
OLMo-2-7B SFT	fewshot	0.982	0.756	0.308	0.551	+0.243
OLMo-2-7B DPO	fewshot	0.986	0.882	0.218	0.536	+0.318
OLMo-2-7B Instruct	fewshot	0.992	0.896	0.159	0.508	+0.349
OLMo-2-7B Instruct	chat	0.985	0.636	0.475	0.552	+0.078
Llama-3.1-8B base	fewshot	0.961	0.623	0.431	0.531	+0.099
Llama-3.1-8B Instruct	fewshot	0.983	0.782	0.444	0.478	+0.034
Llama-3.1-8B Instruct	chat	0.999	0.804	0.589	0.538	−0.051
<i>claude-haiku-4-5</i> (Phase 2)	sampled	—	1.000	0.398	0.096	−0.302

against Haiku’s 1.000 in Phase 2. OLMo base picks the gamble at roughly a coin flip in every frame. The few-shot wrapper is doing what it was built to do, which is to teach the answer format, and it is not producing value-tracking behavior.

This reframes the attribution question the phase was designed to ask. The plan was to test whether post-training bends existing preferences toward or away from human patterns. The base models show there is little to bend: post-training appears to install most of the value-tracking that makes “preferences” a meaningful description of the output at all. A base model at 0.56 dominance compliance does not have unusual preferences, it has no measurable ones, and any structural parameter fit to its choices describes a pattern closer to statistical noise than to taste.

5.3 The staircase: attribution to supervised fine-tuning

The OLMo-2 stages are the core of the phase. All four checkpoints are elicited under the identical few-shot format, which is OLMo’s well-behaved format (Section 5.4), so the format confound is held fixed by construction rather than by argument. Table 2 and Figure 1 report the result.

Model-free: SFT is the largest step, and the later stages keep moving. Dominance compliance rises from 0.555 to 0.756 at SFT (+0.200, 95% CI [+0.177, +0.226]), then to 0.882 at DPO (+0.126, CI [+0.098, +0.152]) and 0.896 at Instruct (+0.014, CI [+0.003, +0.024]). The frame gap more than doubles at SFT, from +0.099 to +0.243 (+0.144, CI [+0.133, +0.155]), and rises to +0.318 at DPO (+0.075, CI [+0.069, +0.082]) and +0.349 at Instruct (+0.031, CI [+0.025, +0.036]). SFT carries about three fifths of the base-to-Instruct change in each. These differences are paired over identical cells and bootstrapped over the 40 gamble identifiers; because the logit readout is deterministic, the intervals reflect only the design distribution, which is why they are narrow.

The framing response is not loss aversion. A positive frame gap means the models gamble *more* when the same lottery is described from a starting endowment, with its downside called a

Table 2: OLMo-2 training staircase, all elicited under the identical few-shot format ($n = 1,680$ cells per stage; structural fits on the 1,600 core cells under $\phi = 1$). Panel A: model-free quantities. Panel B: the Phase 2 specification, with 95% profile-likelihood intervals (gamble-clustered) for the post-trained stages and wild-cluster-bootstrap intervals for the base model, where the profile falls back to dispersion scaling and is too narrow. Panel C: the same model plus a constant c on the choice index (negative = lean toward the safe option), with bootstrap intervals; the c intervals are somewhat off-center at DPO and Instruct (Section 5.4). Quasi-LR for adding c : 360, 31, 33, 35. [†] = estimate at the edge of the parameter space.

	Base	SFT	DPO	Instruct
<i>A. Model-free</i>				
Dominance	0.555	0.756	0.882	0.896
Frame gap	+0.099	+0.243	+0.318	+0.349
<i>B. No constant (Phase 2 specification)</i>				
λ	0.00 [†] [0, 0.52]	1.14 [1.01, 1.26]	1.08 [0.96, 1.19]	1.06 [0.95, 1.16]
γ	1.84 [1.32, 2.58]	0.38 [0.17, 0.57]	0.42 [0.24, 0.60]	0.35 [0.20, 0.52]
α	1.00 [†] [0.87, 1]	0.06 [0, 0.23]	0.04 [0, 0.20]	0.03 [0, 0.18]
Deviance	120.7	228.8	421.2	457.2
<i>C. With a constant lean c</i>				
c	+0.77 [+0.71, +0.86]	−0.80 [−0.83, −0.51]	−1.02 [−0.97, −0.56]	−1.30 [−1.24, −0.70]
λ	0.00 [†] [0, 0.31]	0.00 [†] [0, 0.76]	0.00 [†] [0, 0.69]	0.00 [†] [0, 0.64]
γ	1.59 [1.17, 1.92]	2.19 [1.05, 6.82]	2.28 [1.31, 5.22]	2.03 [1.24, 3.62]
α	0.32 [0.18, 0.43]	0.34 [0, 0.58]	0.27 [0, 0.47]	0.26 [0.02, 0.45]
Deviance	95.3	224.6	413.2	447.5

loss, which is the opposite of what loss aversion predicts. Pairing each mixed-frame cell with its gain-frame twin and regressing the difference in log-odds on the sizes of the stated gain and the stated loss (gamble-clustered standard errors) shows what drives it. The extra gambling grows with the stated gain at every stage (per log unit: +0.13 base, +0.26 SFT, +0.43 DPO, +0.49 Instruct, standard errors 0.01 to 0.06) and barely responds to the stated loss (−0.11, −0.10, −0.23, −0.23, standard errors 0.03 to 0.15). The mixed-frame wording names the gain explicitly (“a gain of $\$G$ ”), and the post-trained models respond to that number.

Structural: what moves depends on one specification choice. Panel B is the Phase 2 specification. Under it, everything moves at SFT: $\Delta\lambda = +1.14$ (bootstrap CI [+0.60, +1.25]), $\Delta\gamma = -1.46$ ([−2.14, −0.85]), $\Delta\alpha = -0.94$ ([−1.00, −0.73]), and the fit leaves the corner of the parameter space where the base model sits. After SFT the parameters hold nearly still. Paired over the same gambles, whose residuals correlate 0.96 between adjacent stages, the post-SFT changes are detectable but small: $\Delta\lambda = -0.06$ [−0.10, −0.03] and $\Delta\gamma = +0.04$ [+0.04, +0.13] from SFT to DPO, and $\Delta\gamma = -0.07$ [−0.10, −0.03] from DPO to Instruct, each under a tenth of the corresponding SFT step. Treated as independent, as the profile contrasts treat them, every post-SFT interval covers zero (SFT to DPO $\Delta\lambda$ profile CI [−0.23, +0.11], $\Delta\gamma$ [−0.24, +0.34]).

Panel C adds a constant c to the choice index. It improves the fit at every stage, and the structural story changes. Loss aversion goes to zero at every stage, so the $\lambda \approx 1.1$ of Panel B was the model using λ , which acts only in the mixed frame, as a partial constant. The base-to-SFT changes in λ , γ and α all have intervals covering zero, and what moves is c : from a lean toward the gamble at base (+0.77) to a lean toward the safe option after SFT (−0.80), a change of −1.57. The lean keeps deepening after SFT, by −0.22 (paired CI [−0.17, −0.05]) at DPO and −0.28 ([−0.26, −0.15]) at

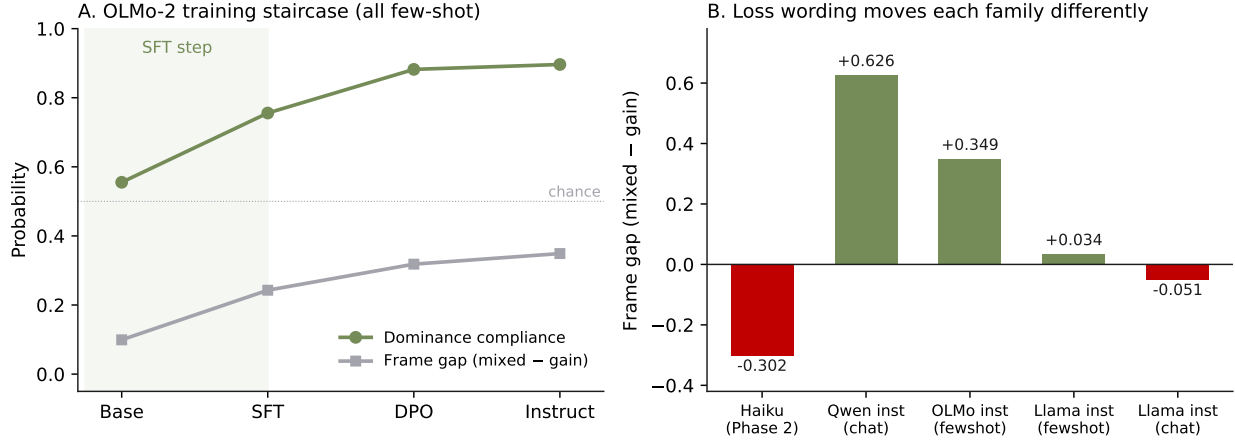


Figure 1: Panel A: the OLMo-2 staircase. Dominance compliance and the frame gap both make their largest move at supervised fine-tuning, and the base model sits near chance on dominance. Panel B: the framing response across families, with the Phase 2 Haiku estimate for comparison. The sign and the magnitude are both family-specific.

Instruct, while the other parameters stay within their intervals. A simulation (Section 5.4) shows that the design separates the two specifications: data generated from Panel B’s fit return λ near one when c is left free, and data generated from Panel C’s fit return λ near zero. The fit comparison therefore favors Panel C, and its reading matches the model-free regression above.

The central claim is stated in three parts, ordered by how much they depend on the model. First, model-free: in this family, on these instruments, SFT is the largest single step in both value-tracking and the framing response, carrying about three fifths of the base-to-Instruct change, and the later stages keep moving both in the same direction. Second, the framing response that SFT introduces is not loss aversion: it rises with the stated gain and barely with the stated loss, and a structural fit that allows a constant finds no weight on losses at any stage. Third, under either specification the structural parameters are essentially in place after SFT; what the later stages add is, under the better-fitting specification, a growing lean toward the safe option. This bears on a literature that has pointed at different stages. Itzhak et al. (2024) found instruction tuning and RLHF amplifying cognitive biases relative to base models; the same group’s later cross-tuning study (Itzhak et al., 2025) finds that bias profiles cluster by pretraining backbone, with finetuning bringing them out. The staircase cannot choose between SFT installing the profile and SFT exposing one planted in pretraining, because the base model fails the dominance control and its preferences cannot be measured (Section 6.1). What it rules out is that the profile first appears at the preference-optimization stage.

5.4 Robustness of the staircase

Interval calibration. The wild cluster bootstrap confirms the gamble-clustered profile intervals for every interior subject: bootstrap intervals are 64% to 103% of the profile widths at the three post-trained OLMo stages and for Llama Instruct. For the five subjects at the edge of the parameter space, where the profile falls back to dispersion scaling, the fallback was too narrow for OLMo base (λ upper limit 0.03 becomes 0.52; γ [1.65, 2.07] becomes [1.32, 2.58]) and roughly right for Llama base. The base-to-SFT contrasts of Panel B still exclude zero with the wider intervals, and so do

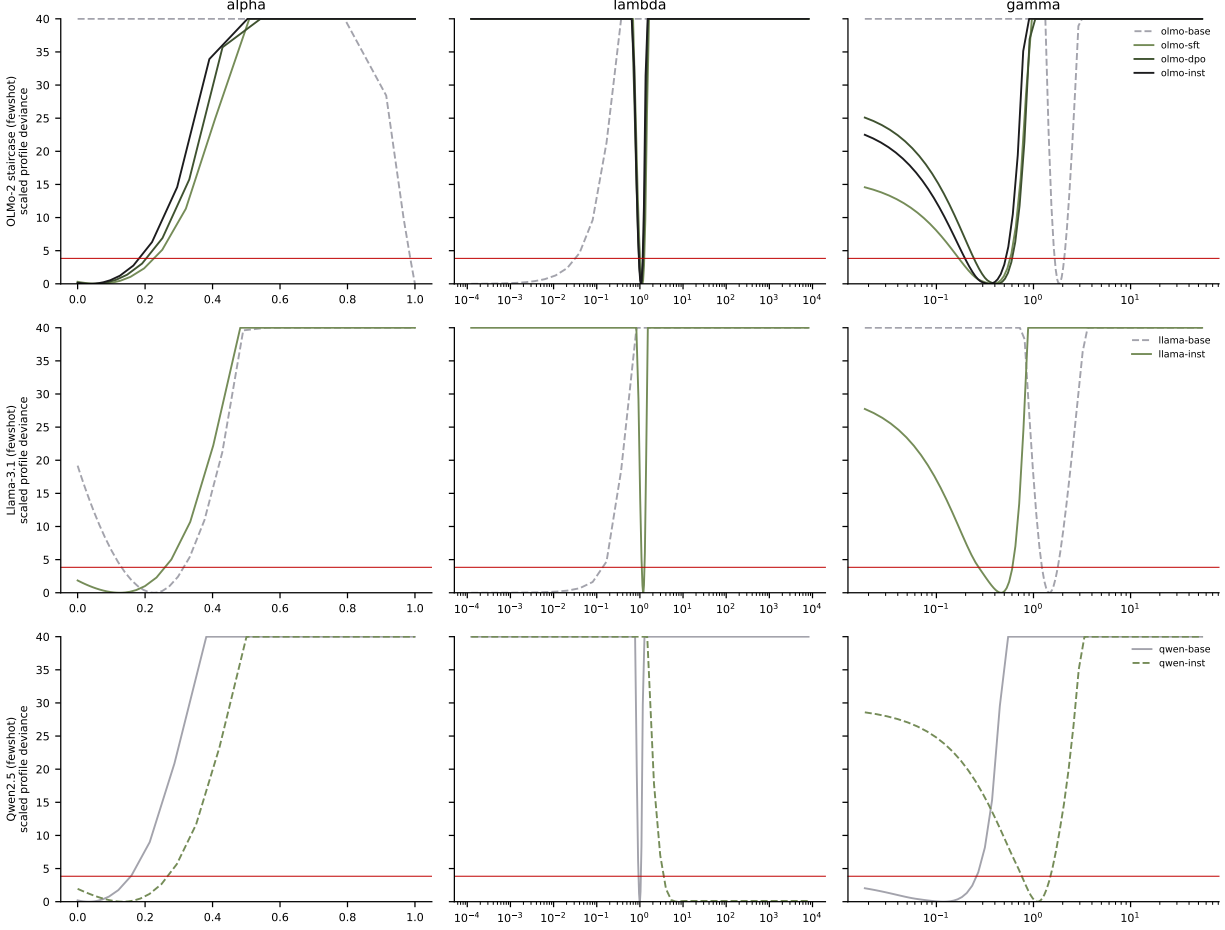


Figure 2: Scaled profile deviance for α , λ and γ under the Panel B specification. The horizontal line is the 95% cutoff; a curve that stays under it to the edge of the axis marks a parameter the data do not bound in that direction. Solid curves use the gamble-clustered scaling, dashed curves the dispersion fallback for fits at the edge of the box. Top row: after SFT the three OLMo stages lie on top of one another, while the base model sits apart.

the Llama base-to-Instruct contrasts in λ and γ . Clipped cells per replicate run 0% to 8% except for Qwen Instruct in few-shot (21%), the near-total refuser already excluded from attribution claims.

The constant. Adding c improves the fit for nine of eleven subjects (not Qwen Instruct few-shot or Llama Instruct chat) and sends λ to zero for all but three (Qwen Instruct few-shot, Llama Instruct chat, and OLMo Instruct chat at 0.07). The bootstrap of the Panel C fit is visibly biased for c at the later OLMo stages, where the point estimates sit just outside their percentile intervals, because the surface has a ridge among c , γ and α and refits drift along it; the signs of the c contrasts are robust, their exact bounds less so. To check identification, `analysis/lossid_power.py` simulates 40 data sets from each of the two SFT fits, with noise shaped like the real residuals (template, gamble and cell components on the logit scale), and refits with λ and c both free. From Panel B’s fit, $\hat{\lambda}$ has median 0.97 (10th–90th percentile [0.45, 1.55]); from Panel C’s fit, median 0.10 ([0.00, 0.40]). The design separates the two because c shifts every frame while λ acts only in the mixed frame. A crossed block that varies gain and loss size independently (`src/design_lossid.py`, 792 cells,

not yet run) narrows $\hat{\lambda}$ only modestly in the same simulation. Its value is the model-free loss-size regression above, which it makes sharper: in the current grid the stated loss spans \$29 to \$186 and correlates with the stated gain (log-log $r = 0.50$), and in the block it spans \$14 to \$271 and is orthogonal to it.

Templates and gambles. Template random intercepts on top of c are significant for every subject and absorb about a fifth of the remaining deviance at each post-trained OLMo stage (20% SFT, 21% DPO, 17% Instruct; 7% at base). The share does not grow at DPO, so wording heterogeneity is not what the later stages add, although its absolute spread does grow: the standard deviation of gamble uptake across the five templates is 0.05 at base and 0.12 to 0.16 after post-training, and one template carries a strong, deepening lean toward the safe option. Gamble random intercepts, with templates as fixed effects, are not significant at any OLMo stage ($\sigma \leq 0.12$, $p \geq 0.18$). Neither kind of heterogeneity moves λ , γ or α once c is in the model.

5.5 Format instability, with a model-specific direction

The dual-format arm was included as a control and returned a result. The same Qwen Instruct weights behave as two different subjects under the two formats. In the chat format it reaches 0.927 dominance compliance and a +0.626 frame gap. In the few-shot format it refuses the gamble almost totally (uptake 0.000 in the gain frame, 0.001 in the mixed frame), the frame gap vanishes, the structural λ has no upper limit (the profile is flat from 3.6 out to the edge of the box, the direction the free- ϕ Phase 2 fit also slid), and $\delta = -4.67$, which says the choices are riding position rather than value.

OLMo Instruct shows the same instability in the opposite direction. Few-shot is its well-behaved format (dominance 0.896, frame gap +0.349); the chat format degrades it (dominance 0.636, frame gap +0.078, $\delta = +3.83$). Llama Instruct is the exception that makes the pattern legible: it is roughly stable across both formats, with dominance 0.782 few-shot and 0.804 chat, and frame gaps of +0.034 and -0.051 . That stability holds only for the model-free quantities. The structural parameters move between Llama’s two formats: $\Delta\gamma = +1.84$ with CI $[+1.17, +3.47]$, $\Delta\lambda = +2.14$ with CI $[+0.89, +4.43]$, and α rises from 0.13 to its upper limit of one. OLMo Instruct’s parameters also move across formats ($\Delta\lambda = +0.31$, CI $[+0.20, +0.43]$; $\Delta\gamma = -0.19$, CI $[-0.37, -0.03]$). Figure 3 puts the three families side by side.

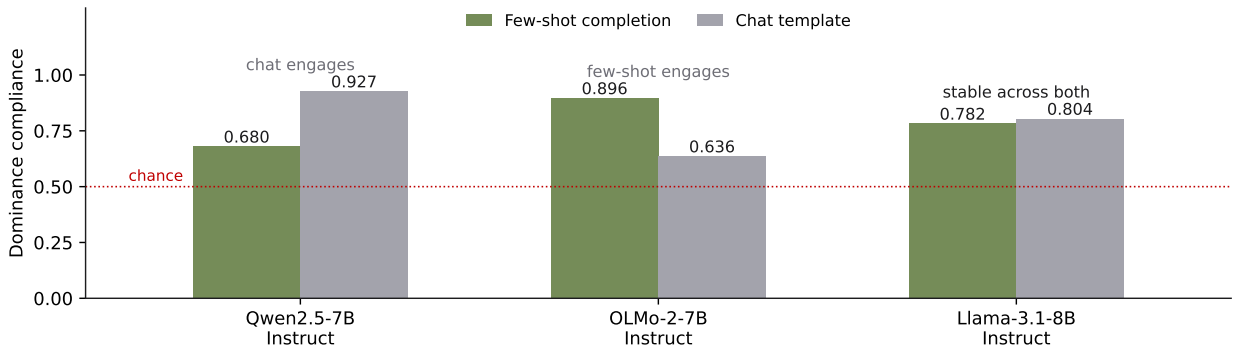


Figure 3: Dominance compliance by elicitation format for the three instruct models. Which format elicits an engaged subject reverses between Qwen and OLMo, and Llama’s dominance compliance is stable across both. The same weights score 0.680 or 0.927 depending only on how the prompt is packaged, so format is part of the measurement rather than a neutral container for it.

Two methodological consequences follow. Format instability generalizes across model families, while which format elicits engaged behavior does not. And a cross-model comparison must hold format fixed *and* verify per model that the fixed format elicits engagement, or it risks comparing an engaged model against one that has disengaged. Dominance compliance is the cheap check, at 80 cells per model per format, but Llama shows it is not a sufficient one: a subject can pass it in both formats and still return different parameters in each. The OLMo staircase in Section 5.3 is all-few-shot, OLMo’s good format, so the staged attribution is unaffected.

The same finding disqualifies one contrast the phase set out to run. The Qwen base-to-instruct attribution under the few-shot format ($\Delta\lambda$ at least $+2.7$ with no upper limit, $\Delta\gamma$ CI $[+0.61, +1.41]$) cannot be read as a post-training effect, because the instruct model’s few-shot behavior does not represent that model in its natural format; it is reported as format-confounded and excluded from the attribution claims. The Llama pair, elicited few-shot on both sides of a format-stable instruct model, gives $\Delta\lambda = +1.20$ with CI $[+1.04, +1.27]$ and $\Delta\gamma = -1.00$ with CI $[-1.35, -0.72]$ from base to instruct (bootstrap CIs $[+1.05, +1.26]$ and $[-1.52, -0.70]$). As with OLMo, the base model’s λ sits at zero, which is the parameter’s natural floor, so the contrast in levels is well defined. As with OLMo too, the λ contrast does not survive a constant term: with c in the model, Llama Instruct’s λ is also zero and the move is in the lean (c from -0.05 to -1.11 , paired CI $[-1.13, -0.94]$). Because Llama’s parameters move with format, this contrast holds for the few-shot format and should not be carried to chat.

5.6 Framing heterogeneity across families

Phase 2 found that Haiku suppresses gambling under loss wording, a frame gap of -0.302 , which runs against the textbook prospect-theory direction of risk-seeking in the domain of losses. Every open model here moves the other way or barely moves. Qwen Instruct in chat amplifies strongly, $+0.626$. The OLMo stages amplify moderately, $+0.24$ to $+0.35$. Llama Instruct is nearly frame-invariant, $+0.034$ few-shot and -0.051 chat.

Three families, three responses to the identical manipulation, and a fourth from the frontier model in Phase 2. The cross-model heterogeneity is a stronger result than any single family’s gap. Loss framing is not a fixed property of language models as a class, and the direction of the response is plausibly a function of the post-training recipe. A literature that estimates a prospect-theory parameter on one instruct model and reports it as a fact about LLMs is over-generalizing from a sample of one recipe. The staircase supplies the complementary within-family fact: framing sensitivity is installed early, then grows monotonically but modestly through the later stages.

5.7 Induced valuation: the models cannot execute an induced rule

Table 3 reports compliance with an explicitly instructed decision rule.

Instruction-only induction fails on cells requiring computation, at every training stage, in both families. Compliance mass on expected-value-clear cells never exceeds roughly 0.55 anywhere, close to indifference. The induction effect grows monotonically up the OLMo staircase (from $+0.017$ to $+0.073$ on all core cells, $+0.029$ to $+0.075$ on clear cells) and stays trivial in absolute terms. Instructed dominance compliance moves the opposite way, climbing to 0.950 and 0.969 in probability mass at the DPO and Instruct stages, which is 1.000 in hard accuracy: every dominant control answered correctly. Figure 4 sets the two against each other.

The instruction helps exactly where the induced optimum requires no arithmetic and does nothing

Table 3: Induced-valuation compliance under a risk-neutral expected-value instruction. Compliance is the mean probability mass on the induced-optimal option, restricted to cells where the induced rule clearly separates the options ($|\ln \text{EU ratio}| \geq 0.05$; 1,221 to 1,240 cells per run). Baseline is the uninduced run from the same model and format, scored against the same rule.

Subject (format)	EV compliance, clear cells			Dominance compliance	
	Baseline	Induced	Difference	Baseline	Induced
Qwen Instruct (chat)	0.524	0.541	+0.017	0.927	0.886
OLMo base (fewshot)	0.516	0.545	+0.029	0.555	0.593
OLMo SFT (fewshot)	0.493	0.525	+0.032	0.756	0.821
OLMo DPO (fewshot)	0.470	0.533	+0.063	0.882	0.950
OLMo Instruct (fewshot)	0.449	0.524	+0.075	0.896	0.969

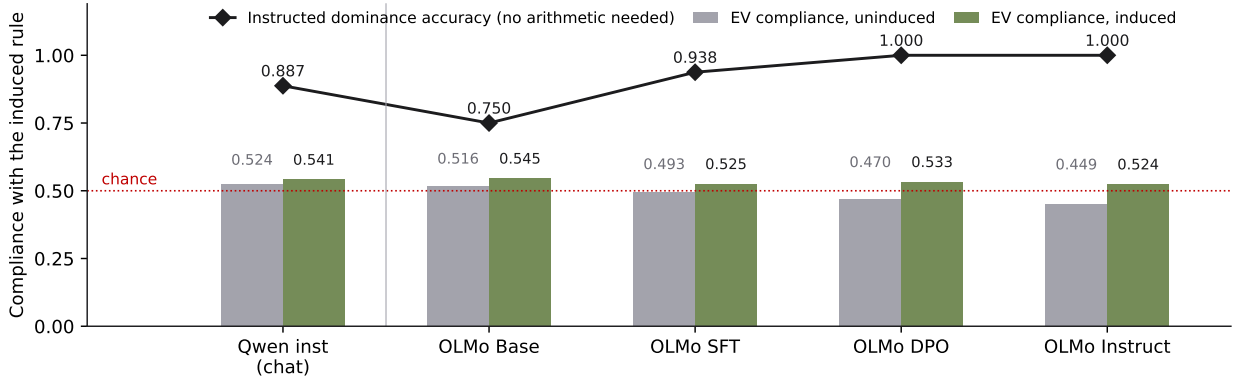


Figure 4: Compliance with an instructed decision rule, by what the rule demands. The bars are expected-value compliance on cells where following the rule requires multiplying p by h : uninduced against induced, both hugging chance at every training stage. The line is accuracy on the dominant controls under the same instruction, where the induced optimum requires no arithmetic, reaching 1.000 by DPO. Post-training buys instruction-following on the cells that need no computation and does not buy the computation.

where it requires multiplying p by h . The sharpest evidence is the square-root run. Told that $u(x) = \sqrt{x}$, Qwen Instruct’s compliance collapsed to 0.124 hard accuracy on clear cells, far below its own uninduced 0.479, and the errors have a legible pattern: gain-frame accuracy of 0.035 corresponds to near-always gambling, which is what comes of comparing square-root-transformed payoffs while dropping the probability term. The model did something with the instruction. It did not do the instruction.

This supports “cannot compute in this format” over “will not comply.” In an immediate-answer elicitation the model must emit a letter at the next token with no intermediate computation available, while post-training steadily improves its following of the no-computation part of an instruction. The induced-value calibration therefore returns a negative result, and the negative result is informative: it puts a ceiling on how much any immediate-answer elicitation of this kind can claim about deliberate optimization. The decisive test is a reason-then-answer arm that lets the model compute the expected value before committing to a letter. Section 5.8 reports it, and it confirms the inability reading.

5.8 Reasoning first: the ceiling was the format

Table 4 and Figure 5 compare the four conditions on the 840 matched cells.

Table 4: The reason-then-answer arm against the immediate-answer runs on the same 840 cells. EV is the share of the 620 clear core cells on which the model puts more than half its probability on the expected-value-optimal option (hard accuracy, so the immediate-answer figures differ slightly from the compliance mass in Table 3). Gap is mean $P(\text{gamble})$ in the mixed frame minus the gain frame. Qwen answers in chat format, the OLMo checkpoints in few-shot.

Condition	Qwen Instruct		OLMo SFT		OLMo DPO		OLMo Instruct	
	EV	Gap	EV	Gap	EV	Gap	EV	Gap
Immediate, uninduced	0.532	+0.626	0.519	+0.241	0.502	+0.319	0.497	+0.346
Immediate, induced	0.569	+0.401	0.545	−0.008	0.549	+0.014	0.542	+0.003
Reason, uninduced	0.829	−0.135	0.681	+0.375	0.741	+0.015	0.740	+0.055
Reason, induced	0.965	−0.145	0.713	+0.358	0.832	−0.071	0.803	−0.024

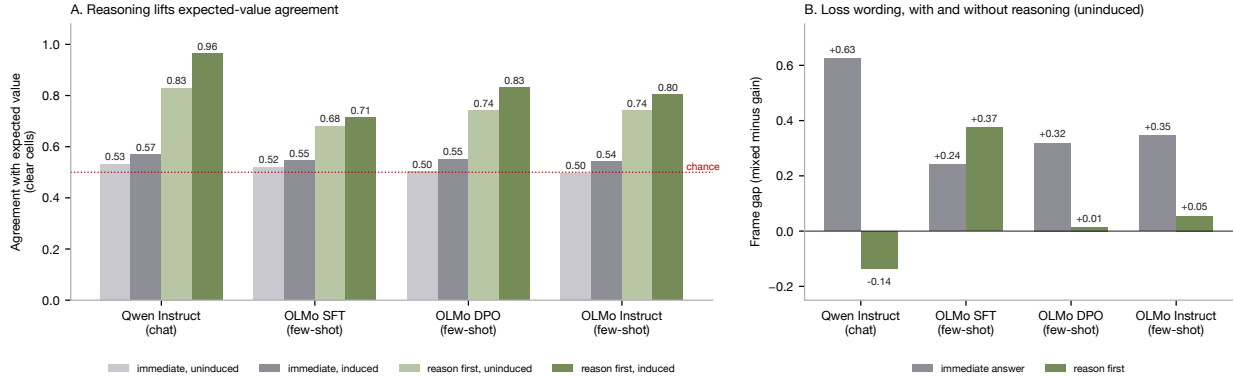


Figure 5: Panel A: agreement with expected value on clear cells, by elicitation regime and instruction. Reasoning moves every subject well above chance, and the instruction adds to it. Panel B: the frame gap without instruction. Reasoning reverses Qwen’s response to loss wording and removes OLMo Instruct’s. OLMo SFT keeps a large gap, which is gone by the DPO checkpoint.

The models can follow the rule when they can write the arithmetic down. With reasoning allowed, induced expected-value agreement on clear cells rises from 0.569 to 0.965 for Qwen Instruct (+0.395, 95% CI [+0.331, +0.456], bootstrapped over the 40 gamble clusters) and from 0.542 to 0.803 for OLMo Instruct (+0.262, CI [+0.199, +0.325]). Dominance compliance reaches 1.000 for Qwen and 0.975 for OLMo Instruct. The traces show the computation directly: “ $(0.60 \times \$234) + (0.40 \times \$0) = \$140.40 \dots \$140.40 > \$139$.” Reasoning moves the model toward expected value even without the instruction. Every uninstructed Qwen trace frames the choice as an expected-value comparison, and agreement rises from 0.532 to 0.829 (+0.297, CI [+0.213, +0.379]). The instruction-only failure of Section 5.7 was a limit of the immediate-answer format, and the structural parameters of Phases 1 through 3 describe that regime: a fast, intuitive choice process, distinct from the deliberate one the same weights can also run.

Under reasoning, the framing effect becomes an accounting error. Qwen’s frame gap reverses from +0.626 to −0.135 (CI [−0.221, −0.053]), the direction Phase 2 found for Haiku (−0.302).

The mechanism is visible in the traces. In the induced run, 21 of the 22 misses on clear cells fall in the mixed frame, and all 21 are cells where expected value favors the gamble: mixed-frame accuracy is 1.000 when the sure amount is optimal and 0.767 when the gamble is. The model computes the gamble as a change from current wealth and compares it with the sure option as a level, as in “ $E_A = (0.47 \times 227) + (0.53 \times 0) - 98 = 8.69 \dots E_B = 98$,” or it nets the gain and loss correctly and then compares the net change with the sure amount. Either way the reference point shifts between the two options, and the shift always favors the safe choice. OLMo Instruct’s gap falls to about zero (+0.055, CI $[-0.075, +0.183]$). Its mixed-frame misses run in both directions and include plain comparison errors (“Since \$151.84 is less than \$117”) that Qwen does not make. Whether Haiku’s anti-textbook Phase 2 response comes from the same mis-referenced computation is a hypothesis these data raise and do not test.

One line of instruction removes the error. If the loss-frame response under reasoning is a bookkeeping habit, naming the right comparison should remove it; if it is a preference, it should survive the correction. We appended one sentence to the prompt (“When you compare the options, compare the final dollar amounts you would end up with under each, not the changes from what you hold now”) and re-ran Qwen Instruct with reasoning on the gain and mixed frames of the same 20 gambles (400 cells). Table 5 reports the result.

Table 5: The clarification check, Qwen Instruct (chat), reasoning first, gain and mixed frames of the 20 matched gambles (400 cells). Mixed EV is agreement with expected value on clear mixed-frame cells; the next column restricts to cells where the gamble is the better option.

Prompt	Gain EV	Mixed EV	Mixed, gamble better	Frame gap
No instruction	0.938	0.787	0.622	−0.135
Clarifying line only	1.000	0.881	0.789	−0.105
Expected-value rule only	1.000	0.869	0.767	−0.145
Rule and clarifying line	1.000	1.000	1.000	0.000

With both the expected-value rule and the clarifying line, all 21 mixed-frame errors disappear and the frame gap is exactly zero. The line alone cuts the errors where the gamble is better from 34 to 19, and the remaining 19 are the same bookkeeping error (“ $EV = 60.48 - 52 = 8.48 \dots EV_B = 52$ ”); none of those traces mentions risk, safety or certainty. Under reasoning, then, the framing response is an arithmetic habit, not a preference, and the model reliably drops the habit when it is told both what to compute and how to compare.

OLMo Instruct does not. The same four prompts on the same 400 cells leave its mixed-frame agreement at 0.62, 0.59, 0.59 and 0.58 (no instruction, line only, rule only, both), and the frame gap within ± 0.06 throughout. The traces show the same slip as Qwen’s and show the line being ignored: after being told to compare final amounts, OLMo writes “a 58% chance to gain \$167 for a total of \$278 $\dots$ The expected value of Arrangement B is $(0.58 \times \$167) + (0.42 \times \$0) = \$96.06$.” The share of mixed-frame traces containing the change-from-holdings number does not fall (0.58 with the rule, 0.59 with rule and line; Qwen’s falls from 0.35 to 0.24 and its errors vanish). About 60% of OLMo’s mixed-frame misses carry that slip, and the rest are comparison errors in both directions. The habit is shared; the ability to act on a correction is a Qwen result.

The written verdict leans toward the safe option, through a stock phrase. Outside the loss frame, Qwen’s uninstructed reasoning still misses 72 of 460 clear cells, and the misses are not risk aversion. In 65 of them the trace computes the gamble’s expected value correctly and then misstates the comparison (“The expected value of Option A is \$152.04, which is less than the

guaranteed \$117”); in 6 it acknowledges the higher expected value and takes the sure amount for its certainty. Checking every numeric comparison between the expected value and the sure amount (`analysis/trace_comparisons.py`) shows that the errors run one way: 40 of the 62 comparisons whose truth favors the gamble are reversed, against none of the 153 whose truth favors the sure amount (Fisher $p < 10^{-27}$). Random slips would reverse both.

A prefill test (`colab/prefill_comparison.py`) locates the mechanism. Feeding each trace back up to the comparison word and reading the model’s probability on “less” against “greater” reproduces every generated word, and shows that 63 of the 65 false verdicts follow one construction, “\$E, which is ... than the guaranteed \$S.” Reached that way, the word is “less” in 63 of 64 cells where the gamble is better; reached any other way, the verdict is wrong in 2 of 149. With the correct number supplied, the phrase still decides: after “The expected value of Option A is \$E, which is” the model puts more than half its probability on “less” in 89% of cells where the gamble is better, and after “Since the expected value of Option A, \$E, is” in 11%. What the wording and the snap answer change is how often the trace uses the construction: 20% of traces under gain wording, 37% neutral and 55% risk, and more often when the immediate answer on the same cell leans safe (clustered logit coefficient on immediate $P(\text{gamble}) -1.28$, $p = 0.004$), with no dependence on how far apart the numbers are ($p = 0.55$). The expected-value instruction removes the false statements in the generated runs (none in 73). OLMo Instruct shows the same one-way asymmetry more weakly (24 of 90 against 5 of 86) alongside two-way comparison errors.

Llama, run through the same arm in chat format, shows that the habit is family-specific. Reasoning lifts its expected-value agreement from 0.515 to 0.821 uninstructed and 0.947 with the rule, close to Qwen, and its frame gap stays near zero in both regimes, as it was in its immediate answers. Its false comparisons run the other way: 41 of 216 comparisons whose truth favors the sure amount are reversed in favor of the gamble, against 1 of 296 in the opposite direction ($p \approx 10^{-15}$), in the template “Since \$115.72 is greater than \$149, Option A has a higher expected value.” The trigger is order, not risk: the false rate is 0.30 when the gamble is listed first and 0.04 when the sure option is ($p \approx 4 \times 10^{-7}$), where Qwen shows no order effect. Llama never uses Qwen’s “which is” construction, and given a one-sentence prefix with the correct number its default verdict is “greater” (“less” 14% of the time after “which is” and 5% after “Since ... is” even when the sure amount is larger). For anyone using reasoning traces to audit a model’s choices, these are measured cases in which the trace reports the right arithmetic and a verdict supplied by a habit, and the habit differs by family: toward the cautious answer in Qwen, toward the first-listed option in Llama.

The staircase depends on the regime. Under immediate answer the structural parameters were essentially in place after SFT, and the later stages added mainly a lean toward the safe option (Section 5.3). Under reasoning, the later stages matter. OLMo SFT keeps a large frame gap (+0.375, CI [+0.286, +0.466]), larger than its own immediate-answer gap, and by the final Instruct checkpoint it is nearly gone: the difference is -0.320 (CI [$-0.449, -0.201$]). Induced expected-value agreement also rises from SFT to Instruct (0.713 to 0.803, difference +0.090, CI [+0.030, +0.160]); the uninduced rise (0.681 to 0.740) is not distinguishable from zero. Running the DPO checkpoint divides the change between the two later stages, and it all belongs to DPO. From SFT to DPO the uninduced frame gap falls by 0.360 (CI [$-0.429, -0.287$]) and induced expected-value agreement rises from 0.713 to 0.832 (+0.119, CI [+0.057, +0.182]). From DPO to Instruct the gap does not move (+0.040, CI [$-0.076, +0.156$]) and induced agreement dips slightly (-0.029 , CI [$-0.054, -0.005$]). Preference optimization leaves the immediate-answer parameters close to where SFT put them and sets how the model reasons; the final stage changes neither.

5.9 Paying for the rule: induced value through training

Smith’s induced-value procedure works because the subject is paid in a currency it wants. A language model has no wealth, but it has an objective its weights are trained on, so a follow-on pilot (Phase 4, `colab/rl_induce.py`) pays the model through training. Each choice resolves as a lottery, and the payoff through the induced utility is the reward in a policy-gradient update of a LoRA adapter, with a penalty on drift from the untrained model. Training uses fresh gambles and only the neutral wording, so the evaluation gambles and the gain and loss framings are held out. The subject is Qwen Instruct in chat, answering immediately; before training, the held-out monitor reproduces its Phase 3 behavior exactly ($P(\text{gamble})$ 0.26 under gain wording, 0.89 under loss wording).

Across six configurations, varying the reward (realized or expected payoff), the baseline, exploration, the reward scale, the learning rate (3×10^{-6} to 10^{-4}) and the balance of the training gambles, payment never taught the model to tell better gambles from worse ones. Held-out balanced accuracy, which is 0.50 for any fixed answer, stayed at about 0.50 in every run through at most 150 steps. What payment did produce was the best available fixed answer: always the sure amount in the first configurations, and always the gamble once the reward structure favored it, since about 62% of gambles favor the risky option and even a count-balanced set leaves the stakes asymmetric. The paid model’s framing gap disappears along with any sensitivity to the options.

The result fits Section 5.8. In a one-letter answer the model does not compute expected value, whether it is told the rule or paid for it, and payment then drives it to a constant policy. This is a pilot with one family, one adapter size and short training, and one fix (gambles balanced by stakes as well as by count) is untested. The more informative incentive experiment pays the model in the reasoning regime, where it already computes; Section 8 describes it.

5.10 Held-out validation

Phase 1 reported honestly that a probability-only logit essentially tied the structural model in sample and beat it on BIC. It does not survive out of sample. Fitting on 30 gambles and predicting cells from the 10 held-out gambles, over 10 random splits, the structural model wins on every split for all three families tested. Mean held-out per-cell deviance is 0.297 for the structural model against 0.401 for the probability-only logit and 0.411 for the constant on Qwen Instruct chat; 0.285 against 0.333 and 0.345 on OLMo Instruct few-shot; and 0.094 against 0.117 and 0.125 on Llama Instruct few-shot. The tally is 10/10 splits in each family (Figure 6), and the ordering is the same on root mean squared error.

The structural model therefore earns its place on prediction, not on interpretability alone. Win probability explains a great deal of the behavior, as Phase 1 found. It does not explain the behavior on gambles the fit has never seen.

6 Interpretation

Post-training creates the subject before it shapes the taste. The intuitive picture of a base model with raw preferences that alignment then bends is not what the data show. Base models fail the dominance control at 0.56 to 0.64 while emitting answer tokens confidently, which is a subject producing a plausible letter rather than comparing two amounts of money. Whatever economic behavior is measurable in the instruct models becomes measurable only during post-

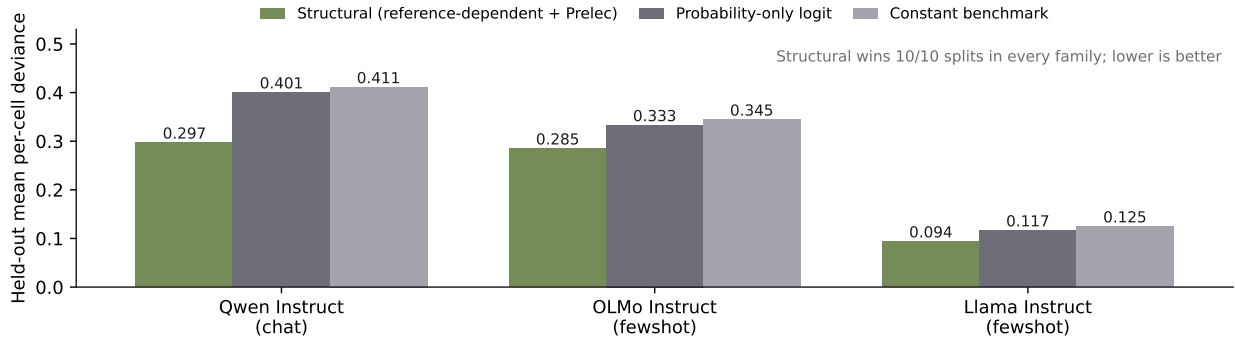


Figure 6: Held-out mean per-cell deviance on gambles the fit never saw, averaged over 10 random 30/10 splits. The structural model beats the probability-only reduced form in every family and on every individual split, and the constant benchmark trails both. Phase 1’s in-sample near-tie between the structural model and the probability-only logit does not survive out of sample.

training. Whether post-training builds it or exposes a disposition laid down in pretraining, the base model’s choices cannot say. Asking where a bias came from presupposes that something was there to be biased.

Supervised fine-tuning comes first, and preference optimization keeps going. Preference-based training is the stage most often blamed for human-like biases, since it is the stage where human judgments enter the weights. In the one family that publishes stage-level checkpoints, the frame gap, the value-tracking, and the interior structural fit are all present after supervised fine-tuning, and the estimated parameters change little afterward. DPO still matters: it adds roughly a third of the total change in both model-free quantities, and in the better-fitting specification it shows up as a deepening lean toward the safe option rather than as shifted preference parameters. So the ordering is clear, and the division of labor is partly clear. SFT does most of the moving; preference optimization amplifies the model-free expression and the safe lean without relocating the other estimates. This is one family with one set of instruments, and the base model’s failed dominance control leaves open whether SFT is the origin or only the point of first visibility. The claim also holds only for the immediate-answer regime: when the model reasons first, the later stages remove most of the frame gap SFT leaves (Section 5.8). Either way, the result points research attention at the least glamorous stage, and away from any single stage as the whole story.

6.1 Installed or revealed? Relation to two recent papers

Two recent papers bear directly on the staircase, and each invites a different objection.

Itzhak et al. (2025): the profile was planted in pretraining. Their cross-tuning design swaps instruction datasets between pretrained backbones and finds that cognitive-bias profiles cluster by backbone, with the instruction data a secondary axis; they conclude that pretraining plants a latent profile and finetuning brings it out. The objection to this paper follows at once: SFT may not *install* the risk and framing profile, only make visible one that was already there. We accept the objection’s premise. The OLMo base model places 0.555 of its probability on the larger of two sure amounts, so the staircase cannot measure the base model’s preferences and cannot tell installation from exposure. The claim this paper makes is the weaker one the design supports: SFT is the first stage at which the profile is measurable, and the structural parameters are in place by then, so

the profile does not originate at the preference-optimization stage. That claim is compatible with planting.

Two features of the data still bear on the planting account, without settling it. First, their bias battery includes the framing effect, the certainty effect and loss aversion, so the two papers measure overlapping behavior with different instruments: theirs are bias scores, compared across finetuned models in a 2×2 cross-tuning of OLMo-7B and T5-11B on Flan and Tulu-2 with three seeds each; ours are structural parameters and terminal-wealth-identical framing contrasts along one training path. Second, the between-family heterogeneity that the planting account locates in the backbone appears in our data only after post-training. The three base models have small frame gaps of the same sign (+0.203 Qwen, +0.099 OLMo, +0.099 Llama). Post-training amplifies the gap in OLMo (+0.349) and Qwen in its working format (+0.626), and shrinks it in Llama (+0.034). Family-specific direction, arriving with post-training, is what an account in which finetuning does more than reveal would predict, though each family’s post-training recipe differs, so backbone and recipe are confounded across families, and base-model frame gaps measured at chance-level engagement carry little weight. The “swayed by finetuning” half of their title has a counterpart here: after SFT, DPO moves both model-free quantities and deepens the safe lean while the other parameters barely move (Section 5.4). And the property that SFT brings out is not the human one: the post-trained models’ framing response tracks the stated gain and not the stated loss, so whatever is planted, it is not loss aversion in the prospect-theory sense.

The decisive test borrows their design. Fine-tuning the OLMo-2 base model on a second instruction mixture, with the instruments and format held fixed, separates the backbone’s contribution from the SFT data’s: if the frame gap and the parameters land where OLMo SFT’s do regardless of the mixture, the profile was planted; if they track the mixture, SFT installs it. A base-model elicitation that passes the dominance control (dominance items among the few-shot exemplars, or more exemplars) is the cheaper complement, since it would let the base model’s own frame gap be read at full engagement.

Wang et al. (2025): base models fit no utility model. On the same Llama-3.1-8B and Qwen2.5-7B pairs, they fit ten utility families (from linear and CRRA through prospect theory and Epstein-Zin) to sampled lottery choices and report that the best-fitting model predicts 69.6% (Llama) and 74.9% (Qwen) of base-model choices, against 93.9% and 82.7% for the instruction-tuned models. They also show that SFT and, more effectively, DPO on choices generated by a target utility move a model toward that target. The objection is that the base-versus-instruct contrast has been done. It has, and on that contrast this paper is a replication, with a deterministic logit readout in place of text sampled at temperature 0.7. What the replication adds is the reason the base fits fail. Dominance compliance of 0.62 (Llama) and 0.64 (Qwen) says that the base models are not comparing amounts at all, so their poor fit is a failure of engagement, not an unusual utility function, and no richer utility class will repair it.

Three further results go beyond their contrast. First, format. Their base models receive a plain-text prompt with in-context examples and their instruct models a chat-formatted prompt, so stage and format change together. Section 5.4 shows that format alone can turn the same Qwen Instruct weights from a near-total refuser into a coherent subject, in a direction that reverses between families; the staircase holds format fixed by construction, and the Llama pair here is compared within the few-shot format. Second, location. Base-versus-instruct compares the two ends of an undisclosed pipeline; the staircase places the change within one public training path. Third, their modulation result is not in tension with the flat parameters after SFT. Their SFT and DPO data are choices generated by a target utility function; OLMo’s preference data are general-purpose.

That DPO *can* move risk preferences when the data are built to do so does not imply that general-purpose DPO *does*, and on OLMo it moved the parameters by less than a tenth of the SFT step.

The bias is a property of a recipe rather than of a technology. Four systems, four responses to the same loss-wording manipulation, from strong suppression through strong amplification to near indifference. The published work reporting prospect-theory parameters for a single instruction-tuned model, including the closest incumbents (Liu et al., 2025; arXiv:2508.08992), is describing a training recipe. Heterogeneity of this size means an evaluation performed on one model transfers to another only by assumption.

Format is a measurement instrument, and it is not neutral. The same weights produced a coherent economic subject and a near-total gambling refusal depending only on whether the prompt arrived as a chat message or a raw completion, and which format worked reversed between families. Any benchmark comparing models under one fixed format is running an uncontrolled interaction.

The estimand is a regime. In an immediate-answer format the models cannot follow an instructed arithmetic rule; allowed to reason, Qwen follows it on 96.5% of clear cells. Every structural parameter in Phases 1 through 3 is therefore a parameter of an immediate-answer choice process. The same weights also run a deliberate process with different properties: it tracks expected value, it responds to loss wording in the opposite direction for Qwen and hardly at all for OLMo Instruct, and it is shaped by training stages the immediate-answer parameters do not register. Its loss-frame response is an arithmetic habit that one sentence of instruction removes. Reporting a model’s risk preferences without the elicitation regime omits half of the result. Phase 2’s free- ϕ degeneracy also reappeared here, as Qwen’s few-shot λ running without limit toward the top of its range, the same direction on the same likelihood surface reached by a different route, which marks it as a property of the specification rather than of any one subject.

7 Policy Relevance

The first audit point is the checkpoint after supervised fine-tuning. Whether SFT installs a behavioral property or first makes it measurable, the checkpoint that follows it is the earliest place the property can be inspected, and in this family it is where most of the change happens. An audit that begins at the preference-optimization stage, which receives most of the governance attention, starts after most of the change has happened. Where to correct a property is a separate question this design does not answer: if the disposition is planted in pretraining, changing the supervised data may move its expression without removing it. The method generalizes: any lab retaining intermediate checkpoints can run this diagnostic on any behavioral property a fixed instrument can measure.

Stage-level attribution requires open checkpoints, which is a policy variable. The result exists because one organization published four checkpoints along its training path. No frontier developer does. An evaluation regime that wants to attribute model behavior to training decisions, rather than only document that the behavior exists, depends on staged-checkpoint access for auditors. That access is a policy choice available now, and it is inexpensive relative to what it buys.

Evaluations need engagement checks or they will report randomness as preference. Two results here are ways a plausible-looking evaluation goes wrong. A base model returns confident answer tokens and coherent-looking parameters while failing to prefer \$70 to \$50. An instruct model returns a completely different economic profile under a different prompt format. Both produce

publishable-looking numbers, and 80 dominance controls catch both. Any evaluation regime for agentic systems should require the engagement diagnostic alongside the headline, for the same reason a survey reports its response rate.

Do not port a measured bias across models. An agent that suppresses risk-taking under loss wording and one that amplifies it will move in opposite directions on the same procurement menu, the same insurance framing, the same triage prompt. A framing vulnerability documented on one model tells a procurement officer nothing reliable about the model deployed, so the measurement must be repeated per model. That is the argument for cheap standardized instruments of the kind this project builds.

State the elicitation regime. The same model reverses its response to loss wording depending on whether it answers immediately or reasons first. Agentic deployments increasingly let models reason before acting, while most behavioral evaluations still read an immediate answer. An evaluation that does not report which regime it measured cannot be compared with one that does, and a framing vulnerability found in one regime may be absent, or reversed, in the regime that is deployed.

Scope. These are properties of specific open-weight checkpoints under a deterministic logit readout with unincentivized forced choice. They are not claimed as universal traits of language models, and the framing-heterogeneity result is precisely the reason such claims are unsafe.

8 Limitations and Next Phase

Loss aversion depends on a constant the Phase 2 specification omitted. The λ estimates of Phases 2 and 3 were fit without a constant lean. With one, λ goes to zero for most subjects here, and the fit improves (Section 5.4). The Phase 2 Haiku estimate ($\lambda = 3.23$), fit to sampled choices with a negative frame gap, should be re-estimated the same way before it is cited. The crossed gain-by-loss block (`data/phase3_lossid_grid.csv`) is ready to run and would sharpen the model-free loss-size test.

Intervals at the edge of the parameter space. Five subjects have a parameter at the edge of the wide box. The gamble-clustered bootstrap gives them calibrated intervals, wider than the dispersion fallback for OLMo base, but at a boundary the bootstrap is itself not exact. The staircase’s model-free claims do not rest on these intervals.

The base model cannot be measured, so origin is open. SFT is the first stage at which the profile is measurable, and that is all the staircase can say about its origin. Two follow-ups would separate installation from exposure: a base-model elicitation that passes the dominance control (more exemplars, or dominance items among the few-shot examples), after which the frame gap can be remeasured; and a probe for a framing direction in base-model activations. The second connects to the interpretability leads in `docs/`.

What DPO changes is only partly modeled. Preference optimization moves both model-free quantities while λ , γ and α barely move and the fit deteriorates. The position parameter is not the channel (+1.18, +1.07, +1.06), and template and gamble random effects absorb a stable share of the deviance at every stage rather than a growing one. A constant lean toward the safe option accounts for part of it. A noise process that is not Fechner, or template-specific noise scales, are the next candidates.

One instruction phrasing in the induced arm. Each induced rule was tested under a single phrasing. A compliance failure under one phrasing is weaker evidence than a failure under several, and the robustness check is unrun.

The reasoning arm is partial. It covers half the gambles, one instruction phrasing, greedy decoding (one trace per cell), and five subjects from three families; Llama has no immediate-answer run under the instruction, so its instruction effect is measured only under reasoning. The trace coding relies on automatic extraction of comparisons written with both numbers (about a quarter of Qwen’s traces), spot-checked by hand. The few-shot reasoning exemplars demonstrate the expected-value arithmetic, which primes the uninduced OLMo runs toward it. The reference-point error is identified from traces read by hand and a numeric text check (the change-from-holdings number within 2%), not a full coding of the reference point each trace uses.

No choice here is incentivized. Every elicitation in Phases 1 through 3 is hypothetical, and the induced-value arm is instruction only. Paid training (Section 5.9) is the closest available analogue to payment, and in the immediate-answer regime it failed to teach the computation. The next incentive experiment pays the model in the reasoning regime, where it already computes, and targets two questions instructions could not settle: whether payment can induce a specified risk-averse utility ($u(x) = \sqrt{x}$, where instruction alone collapsed to 0.124 compliance), and whether paid reasoning carries over to immediate answers. It needs generation during training, roughly ten to twenty times the compute of the pilot.

Lottery choices are not advice. Every subject here chooses lotteries for itself, and deployed models advise others on decisions in other units. Whether the measured curvature, loss aversion, wording effect and verdict habits predict the risk posture of a model’s recommendations is untested. A follow-on (`docs/extension-advice-2026-10.md`) maps the same gambles, number for number, into advisory questions in national security domains (lives protected, objectives held, crisis escalation), posed to the model as itself and as an adviser to a commander or an official, in both frames and both option orders; it then moves to open-ended scenario briefs coded for risk posture. The test is whether money-domain behavior predicts advisory behavior cell by cell and whether the parameters are invariant across domains and roles.

Hardware, quantization, and one staircase. All runs use 4-bit quantization on free Colab T4 hardware. The exercise compares checkpoints under identical conditions, so a small uniform quantization effect would not disturb the staircase, though this is unverified against full precision. The stage-level attribution also rests on one family, because OLMo-2 is the only family publishing intermediate checkpoints. A second staged family would turn the central claim into a replication.

Phase 3 has done its job. It carries the Phase 1 and Phase 2 instruments to models whose training history is visible, shows that the measured profile is in place at supervised fine-tuning before preference optimization begins, shows that the framing response it installs or reveals is not loss aversion, shows that base models lack the value-tracking that would make the word “preference” apply, documents that the framing response differs in sign across families, and answers the Phase 1 near-tie out of sample. It also found the ceiling of its own elicitation and, with the reasoning arm, raised it: the models can follow a stated rule when they may compute, their framing response and training-stage profile both depend on whether they do, the framing response that remains under reasoning is a bookkeeping habit one sentence removes in Qwen and not in OLMo, and the written reasoning can attach a habitual verdict to correct arithmetic.

References

Donald W. K. Andrews. Inconsistency of the bootstrap when a parameter is on the boundary of the parameter space. *Econometrica*, 68(2):399–405, 2000.

- arXiv:2508.08992. Cumulative prospect theory parameters of instruction-tuned language models. arXiv:2508.08992, 2025. Cited in the drafts as “CPT estimation on LLMs”.
- A. Colin Cameron, Jonah B. Gelbach, and Douglas L. Miller. Bootstrap-based improvements for inference with clustered errors. *Review of Economics and Statistics*, 90(3):414–427, 2008. doi: 10.1162/rest.90.3.414.
- Stefano DellaVigna. Structural behavioral economics. In B. Douglas Bernheim, Stefano DellaVigna, and David Laibson, editors, *Handbook of Behavioral Economics: Applications and Foundations 1*, volume 1, pages 613–723. North-Holland, 2018. NBER Working Paper 24797.
- Charles A. Holt and Susan K. Laury. Risk aversion and incentive effects. *American Economic Review*, 92(5):1644–1655, 2002. doi: 10.1257/000282802762024700.
- Itay Itzhak, Gabriel Stanovsky, Nir Rosenfeld, and Yonatan Belinkov. Instructed to bias: Instruction-tuned language models exhibit emergent cognitive bias. *Transactions of the Association for Computational Linguistics*, 12, 2024. arXiv:2308.00225.
- Itay Itzhak, Yonatan Belinkov, and Gabriel Stanovsky. Planted in pretraining, swayed by finetuning: A case study on the origins of cognitive biases in LLMs. In *Conference on Language Modeling (COLM)*, 2025. arXiv:2507.07186.
- Daniel Kahneman and Amos Tversky. Prospect theory: An analysis of decision under risk. *Econometrica*, 47(2):263–291, 1979. doi: 10.2307/1914185.
- Botond Köszegi and Matthew Rabin. A model of reference-dependent preferences. *Quarterly Journal of Economics*, 121(4):1133–1165, 2006. doi: 10.1093/qje/121.4.1133.
- J. Liu, Y. Yang, and K. Y. Tam. Evaluating and aligning human economic risk preferences in LLMs. In *Proceedings of the 2025 Conference on Empirical Methods in Natural Language Processing (EMNLP)*, 2025. arXiv:2503.06646.
- Leslie E. Papke and Jeffrey M. Wooldridge. Econometric methods for fractional response variables with an application to 401(k) plan participation rates. *Journal of Applied Econometrics*, 11(6):619–632, 1996.
- Drazen Prelec. The probability weighting function. *Econometrica*, 66(3):497–527, 1998. doi: 10.2307/2998573.
- Alex Rees-Jones and Ke Wang. An approach to testing reference points. NBER Working Paper 30773, National Bureau of Economic Research, 2022.
- Steven G. Self and Kung-Yee Liang. Asymptotic properties of maximum likelihood estimators and likelihood ratio tests under nonstandard conditions. *Journal of the American Statistical Association*, 82(398):605–610, 1987.
- Vernon L. Smith. Experimental economics: Induced value theory. *American Economic Review*, 66(2):274–279, 1976.
- Kenneth E. Train. *Discrete Choice Methods with Simulation*. Cambridge University Press, 2nd edition, 2009.
- Amos Tversky and Daniel Kahneman. The framing of decisions and the psychology of choice. *Science*, 211(4481):453–458, 1981. doi: 10.1126/science.7455683.
- Amos Tversky and Daniel Kahneman. Advances in prospect theory: Cumulative representation of uncertainty. *Journal of Risk and Uncertainty*, 5(4):297–323, 1992. doi: 10.1007/BF00122574.
- Yikai Wang, Xiaocheng Li, and Guanting Chen. Risk profiling and modulation for LLMs. arXiv:2509.23058, 2025.
- George Wu and Richard Gonzalez. Curvature of the probability weighting function. *Management Science*, 42(12):1676–1690, 1996. doi: 10.1287/mnsc.42.12.1676.